United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Final Exam Trimester: Spring 2025 Marks: 40 Time: 2 Hours

Code: CSE 3411 Course Title: System Analysis & Design

"Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules."

Answer all the following questions

Question 1: [CO3] [18]

Consider the following scenario:

"**UCAM**" is the academic management software of United International University, designed to handle operations for ease of management of academic programs. All users log in through the system and have personalized dashboards. The system contains 3 types of primary **Users**: **Students, Faculty Members, and Administrators**.

A **Student** logs in to view their profile, check attendance, see academic results, and handle financial transactions. During the pre-advising phase, students can register for available **courses.** Each Course has multiple **Sections**. A Section is composed of a **course, a schedule, and a list of enrolled students.** At the beginning of the semester, administrators at United International University log in to the UCAM system using their credentials. Once authenticated, they can open a course for the upcoming term by specifying details such as course code, title, number of sections, capacity, and schedule. UCAM registers the course as available for pre-advising, making it visible to eligible students. Meanwhile, a student logs into UCAM and navigates to the pre-advising portal. There, the student browses available courses based on their program, completed prerequisites, and semester plan. Upon selecting a desired course, the student views the list of available sections created by administrators, including schedules and seat availability.

Faculty members use UCAM to open courses for pre-advising, manage their sections, take attendance, and submit grades. Each faculty member is associated with one or more sections. They can only perform grading or attendance actions when a section is in an "Active" state.

Administrators have overarching control. They manage the academic calendar, configure course offerings, approve new users, and ensure system-wide access control.

Payments are tightly coupled to **student accounts** and become mandatory before final course registration. Once payment is confirmed, the student enters a "Registered" state. Any failure in payment reverts the student to a "Pending" state.

a) Mention the major guidelines/rules to determine the use case in any software system. Design a Use Case Diagram for the "UCAM" application. Ensure that you include at least one include or extend relationship. [1+4]

b) Mention some common rules/guidelines to identify the classes in any software system. Draw the Class Diagram and CRC card for the "UCAM" application by identifying the main classes and their relationships. [1+4]
c) Create a State Diagram that depicts the life cycle of a **Student** in the "UCAM" system. [4]
d) Draw a Sequence Diagram that demonstrates the process of a student for selecting a course and a section at the beginning of the semester. [4]

Question 2: [CO4] [6]

Imagine you and your team are running a CNC machining firm that specializes in precision parts for the renewable energy sector. To expand operations, your firm invested $20,000 initially in advanced CNC machines. In the third year, an additional $3,000 was spent on equipment upgrades and installation. Later, in the fifth year, you incurred $5,000 in maintenance costs for the machines to ensure optimal performance. The revenues from machining contracts were recorded as follows: $40,000 in Year 1, $55,000 in Year 2, $70,000 in Year 3, and $80,000 in Year 4. By the end of Year 5, your accounting shows a cumulative cash flow (total net profit after deducting all expenses from revenues) of $2,000.

a) Using the NPV method, calculate the earned revenue in Year 5. [2]
b) CNC machining is still emerging in Bangladesh's industrial landscape, where manual processes dominate. Analyze the feasibility of a CNC machining firm in the context of Bangladesh using feasibility analysis techniques. [4]

Question 3: [CO3] [11]

a) "An effective Software Requirements Specification (SRS) must uphold at least two fundamental properties – correctness and completeness. Define and justify. [2]
b) Determine the functional and non-functional features/requirements for the scenario mentioned in Question 1. [4]
c) Mention some (at least three) best practices/guidelines to do the UI design for any Software product. Draw a UI Design for the personalized “Dashboard” of a student in the UCAM application. Mention the UI design principles that you have followed for this case. [3+2]

Question 4: [CO3] [5]

Write down an SRS document and present it as a group having the following parts on your software project work covering: Introduction, System Study, System Analysis, Feature list fixation with Functional and Non-functional requirements, Different Feasibility Analysis, System design (UML diagrams), UI design through Figma software, implementation plan etc. (**no need to answer this question in the examination**, it has already been evaluated in classroom activities). **[5]**